CAPSTONE FINAL REPORT

Design and Development of a Time-limited Shop 2D Survival Combat Shooter

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Abstract

This project designed and developed a browser-based 2D survival shooter to investigate how scarcity cues might be integrated into a playable game environment through a time-limited shop. The prototype was built around a stage-based gameplay loop that included battle, rewards, progression, and purchasing options. Players must fight enemy waves, acquire scrap, upgrade their aircraft, and prepare for later stages while under growing strain.

The prototype's key design feature is a two-condition shop structure. The first store is untimed, but subsequent shop visits have a noticeable 15-second countdown. This concept differentiates between low-pressure and time-limited buying circumstances within the same gameplay system. The prototype also separates score and scrap, features many enemy tiers and boss encounters, and connects shop purchases to subsequent battle survivability.

The final prototype successfully implemented the desired scarcity mechanism as a useful gameplay component rather than a simple visual interface element. The findings show that time-limited purchase pressure can be integrated into a complete survival shooter loop in which battle danger, limited resources, and stage progression influence decision-making. Although the project does not include formal behavioural testing, it does provide a solid design artefact for evaluating timed scarcity in games and a foundation for future empirical research into player buying behaviour under time constraints.

Keywords

scarcity cues; time-limited shop; purchase intention; virtual goods; browser-based game; survival shooter

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1. Introduction

1.1 Background

Digital commerce and game monetization both heavily rely on scarcity cues. Urgency signals, like countdown timers and limited-time offers, are used to speed up decisions and increase purchase intent. Prior research has demonstrated that while limited-time scarcity can promote impulsive purchases because of urgency and arousal, scarcity messaging can increase buy intention. Virtual product research indicates that platform and service design, in addition to the item itself, have an impact on gaming sales. However, game monetization design has come under fire for pushing players to spend through manipulative practices rather than informed choice.

This topic matters because game spending now sits inside a large and growing market. Small interface choices such as timers, colour changes, and purchase windows can influence player behaviour during play. Although much prior work on scarcity appeals comes from retail, e-commerce, and live commerce, games provide a different setting. Players make choices while managing attention, risk, reward, and ongoing action, which makes scarcity cues in games a design issue as well as a consumer behaviour issue (Hamari & Keronen, 2017).

1.2 Project Context

This project addresses that gap by developing a browser-based 2D survival shooter with Phaser. During shop phases, which are interspersed between the game's recurring battle rounds, players can spend in-game currency on upgrades and survival items. After engaging adversaries, gaining scrap, visiting a store, and upgrading the aircraft, the player advances to the next step. This loop provides a controlled setting for studying the possible impacts of scarcity cues on play-related purchasing behaviour.

The time-limited shop system is a fundamental component of the concept. Later shop visits operate under a 15-second timer, but the first shop opens without a countdown. Before the shop closes, the player must choose how to spend the limited currency on things like better missiles, faster fire, shield defence, and health recovery. Additionally, the game distinguishes between score and scrap so that purchasing power and gameplay performance stay separate. Store decisions have apparent implications for subsequent gameplay due to enemy tiers, boss stages, health management, and revival logic.

1.3 Aim and Objectives

The project's purpose is to create a playable 2D survival shooter that uses scarcity cues in an in-game store, as well as to research how the design may influence players' inclinations to purchase while playing.

The initiative has five goals, which are:

1. Use Phaser to construct and develop a browser-based 2D aerial combat game.
2. Design a gameplay loop that combines shop interaction, resource collecting, combat, and stage progression.
3. Implement a time-limited shopping system with a limited purchase window and a visible countdown.
4. Integrate upgrade, health, shield, adversary, and boss systems such that shop choices impact future survival.
5. Evaluate the prototype as a design artefact and explore how timed scarcity indicators can

impact player purchasing behaviour in a game scenario.

1.4 Scope and Limitations

This project consists solely of the concept, production, and analysis of a single-player browser game prototype. The study focuses on the game's functional systems, the reasoning behind the time-limited shop, and the design justification for the scarcity cue. It makes no claim to provide a thorough behavioural experiment with large-scale testing, formal participant recruitment, or statistical validation of player responses.

The prototype has certain practical restrictions as well. There is only one main game mode, a handful of shop products, and a limited number of enemy types. Online play, matchmaking, account systems, long-term player data monitoring, and a commercial content pipeline are all excluded. Because the project's purpose is to build and test a functional prototype rather than a finished commercial product, these constraints are acceptable.

1.5 Report Structure

The report is broken into six sections. Chapter 1 describes the project's background, goal, and scope. Chapter 2 examines the relevant literature. Chapter 3 describes the design and development technique. The findings are presented and analysed in Chapters 4 and 5, and the report concludes with Chapter 6.

2. Literature Review

2.1 Browser-Based 2D Game Development

Browser-based 2D game development is ideal for prototype-driven capstone projects since it allows for rapid iteration, direct deployment, and modular implementation. Phaser is a popular HTML5 framework for 2D game creation that includes support for scenes, physics, asset management, and browser rendering. This characteristic makes it perfect for a compact game prototype that requires integrating battle logic, interface elements, and time-limited purchase mechanics in a single web-based system (Phaser Studio, n.d.). The browser structure used in this project allows for the analysis of scarcity cues in a playable environment rather than a static mock-up. It enables action, interface feedback, and incentive systems to function together in a single controlled prototype.

2.2 Core Gameplay Loops in Arcade Shooter Design

Short repeating loops consisting of threat, action, reward, and progression are commonly employed in arcade shooter design. The player must make numerous tactical decisions under pressure, deal with growing difficulties, and receive immediate feedback. Because it maintains a clear link between skill, survival, and reward, this structure remains effective. The idea that recurrent challenge-response loops can maintain engagement when goals and feedback remain explicit is further supported by research on gameful systems and flow (Oliveira et al., 2025).

This study applies similar reasoning to a stage-based survival shooter, with shop times transforming performance into decision-making and combat phases introducing risk and reward. As a result, rather than being a separate store action, the decision to purchase becomes a component of the gaming cycle.

2.3 Player Decision-Making Under Time Pressure

Scarcity cues can increase purchase intention by raising urgency and diminishing deliberation, according to a large body of evidence. Wu et al. (2021) showed that scarcity promotion can increase impulse purchases via arousal-based mechanisms, whereas Aggarwal et al. (2011) revealed that consumer reaction to scarcity signals is dependent on how scarcity is presented.

Recent study supports similar patterns. Nyrhinen et al. (2024) demonstrated that digital persuasion settings and self-control influence online impulse purchases, whereas Kathuria and Bakshi (2025) discovered that limited-time promotions have a positive impact. All of these studies suggest that time restrictions may induce consumers to make impulsive, emotionally laden purchases rather than planned ones.

This project uses that idea directly through a 15-second countdown in later shop visits, turning the timer into a gameplay mechanic rather than a decorative feature.

2.4 Reward Systems, Upgrade Economies, and Player Motivation

Research on virtual items shows that factors other than item usability influence gamers' purchase inclinations. According to Hamari and Keronen's (2017) research, virtual goods purchases relate to social influence, attitude, and service-related design aspects. Tyrväinen and Karjaluoto (2024) discovered that desire to pay in freemium settings depends on functional, hedonic, social, and price value. Serafim et al. (2022) also found that attitude and perceived value were key determinants of purchase intention for virtual objects in free games.

This is significant for the current study since the prototype store sells functional upgrades instead of cosmetics. As a result, purchasing intention is influenced by perceived upgrade value, predicted future difficulty, and available resources. This strategy is further reinforced by the distinction between score and scrap, which isolates purchasing power from gameplay performance.

2.5 UI Feedback and Urgency Cues in Games

Scarcity effects can be caused by more than just text. The interface's impact is determined by how it expresses urgency through design, look, colour, and timing. One obvious example is a countdown timer. Tuncer et al. (2024) found that in the limited-time scenario, scarcity cues on e-commerce sites increased negative emotions associated with impatience while decreasing perceived friendliness. Furthermore, Koh and Seah (2023) revealed that time-limited messages and other dark patterns had a significant impact on product purchasing decisions.

The same conclusion is obtained from research on dark patterns and deceptive countdown clocks. Countdown-based interfaces should be seen as behavioural design tools rather than neutral visual elements, as illustrated by Tiemessen et al. (2023), Luguri and Strahilevitz (2021), and Mathur et al. (2019).

In this project, the player can view time, scrap, health, shield state, and upgrade status all at once. The countdown is also visible, and the purchasing period is genuinely limited. As a result, the decision space is constricted, and both game logic and interface design create urgency.

2.6 Scarcity, Monetisation, and Ethical Concerns

The study also shows how scarcity-based monetization can generate ethical concerns when it coerces people into making purchases or takes advantage of underlying biases. Salehudin and Alpert (2022) discovered that when monetization is perceived as aggressive or unfair, mobile game players usually resist in-app purchases. Furthermore, Cartwright (2022) argued that monetized game systems, such as loot boxes, can act as aggressive marketing strategies, especially for vulnerable consumers.

This topic has been expanded in recent HCI work. Mann et al. (2024) proposed an ethical framework for microtransactions in mobile games, arguing that creators need more detailed

direction when building commercialised systems. Scarcity cues can operate on a continuum of persuasive and manipulative design, making them relevant in this setting.
This project limits that risk by using earned in-game currency rather than real-money transactions. That choice allows the report to examine the logic of time-limited purchase pressure in a more controlled design setting.

2.7 Research Gap

Literature supports three general conclusions. First, shortage indications usually increase feelings of urgency, enthusiasm, and impulsive buying. Second, bigger system characteristics such as value, fairness, incentive, and monetization structure influence the purchasing of virtual goods. Third, when countdown interfaces and other urgency cues become coercive or misleading, ethical concerns may develop.
The contextual gap is the most significant. Retail, e-commerce, and general digital commerce continue to produce most of the evidence. Placing temporal scarcity within a viable game loop in which the player controls advancements, rewards, and survival in real time takes less effort. This project addresses that gap by developing a functioning 2D survival shooter in which timed scarcity occurs within the play structure rather than outside of it.

3. Method

3.1 Project Methodology and Design Rationale

This project was an engineering design project with the goal of creating a play browser-based gaming prototype. The prototype's goal was to include scarcity indicators into an active gameplay environment and investigate how a time-limited shop influences purchase intention while playing. Rather than showing scarcity in a separate store interface, this concept included timed by pressure into a full survival shooter loop that includes battle, reward, progression, and upgrade options.
The project necessitated a functional interactive system rather than a static interface mock-up; hence a prototype-driven development strategy was chosen. The game was built as a browser-based 2D prototype with JavaScript, HTML, and Phaser. This technical decision was suitable given that the prototype needed to handle scene management, keyboard input, arcade-style physics, interface overlays, and repetitive gameplay testing in a lightweight web context.
Two related concepts served as the foundation for the overall design rationale. First, prior studies indicate that timed scarcity cues may induce a sense of urgency and encourage consumers to make purchases more quickly. Second, because purchases are made in the context of level development, resource scarcity, and survival pressure, this effect is probably going to function differently in games. Because of this, the prototype was created as a stage-based survival shooter where purchase and battle periods alternate. Instead of operating as a separate menu, this structure made it possible for the shop to be integrated into the gaming system.
To meet this structure, the prototype was separated into four main scenes: a preload scene, a menu scene, the main gaming scene, and a specialist user interface scene. The preload scene handles asset loading, the menu scene handles difficulty selection, the gameplay scene handles combat and progression logic, and the user interface scene handles player data, shop interaction, countdown displays, and revive prompts. This modular design was chosen to keep presentation logic and gaming logic apart and to facilitate testing and system improvement. Figure 3.1 depicts the connection between the principal scenes, gameplay entities, and configuration layer,

which helps to illustrate the hierarchical organisation of the prototype.

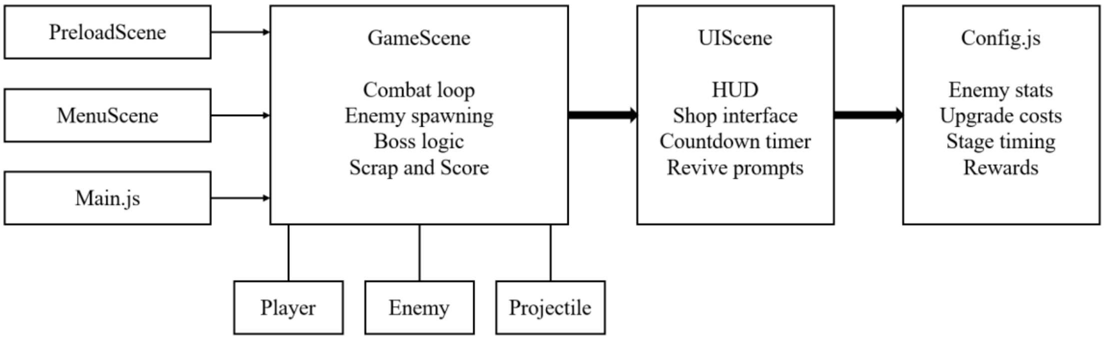


Figure 3.1 Overall system architecture of the prototype

The prototype improves flexibility and allows for future testing and refinement by separating interface management from combat and progression logic, as seen in Figure 3.1. The project employs a recurrent temporal framework at the gameplay level. Following each normal fighting stage, the player enters a 30-second shop phase or, at times, a boss stage. Later store phases have a 15-second time limit and close automatically when the countdown is complete, however the first shop is displayed without a timer. This feature was purposely included in the prototype to allow for comparisons between a low-pressure buying scenario and subsequent planned buy situations. The countdown timer is regarded as the major operational form of the scarcity cue in this study. To clarify this design logic, Figure 3.2 illustrates the full gameplay loop of the prototype, including combat, reward, shop access, and boss progression.

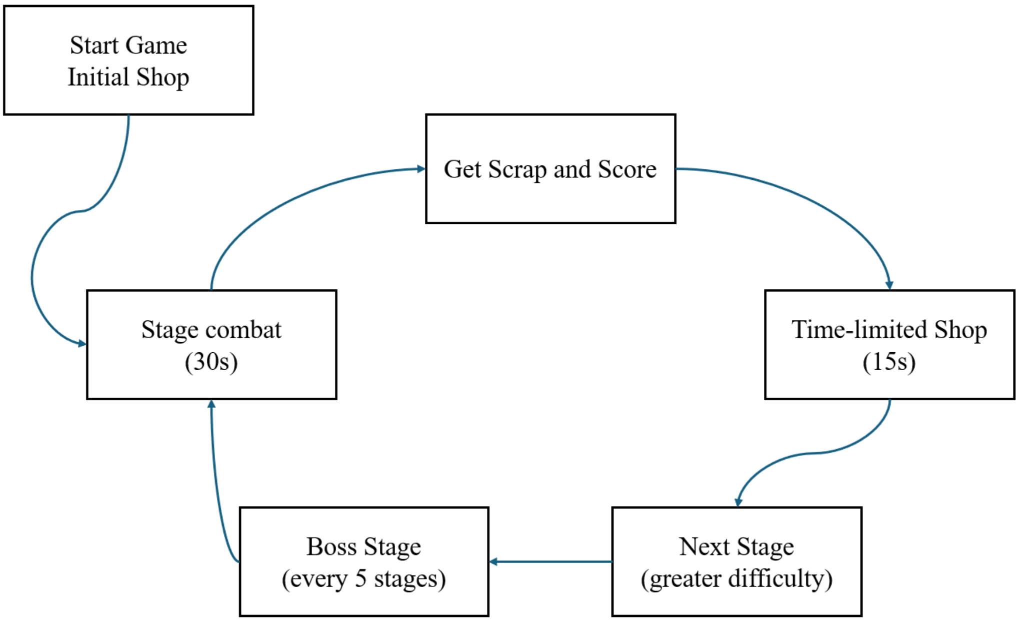


Figure 3.2 Stage-based gameplay loop of the prototype

Additionally, score and in-game currency are separated in the final prototype. Scrap is used as a purchasing resource in the shop, while score monitors player performance and level advancement. This distinction was made since the prototype's purchase demand is based on few resources rather than just score. Even if players are skilled fighters, they still need to use their time wisely when allocating scrap. This layout helps the store become a significant setting for making decisions that are impacted by urgency, available resources, and anticipated future challenges.

This project does not claim to be a thorough behavioural experiment with official participant recruitment or large-sample statistical testing from a methodological perspective. Rather, it provides a functional design artefact that implements scarcity indicators within a gaming context. As a result, the prototype serves as a playable environment for further empirical research and design study.

3.2 Functional Requirements

The prototype was designed to assist the project's goal by connecting gameplay, progression, and scarcity-based purchasing pressure. Rather of being a standalone feature, these specs were designed to include the time-limited shop concept into a comprehensive and unified game system. Table 3.1 summarises the prototype's major functional needs.

Requirement ID	Functional requirement	Purpose in this project
FR1	Player aircraft movement and continuous shooting	To support active combat interaction
FR2	Enemy spawning, wave progression, and boss stages	To create increasing gameplay pressure
FR3	Scrap and score as separate systems	To distinguish purchasing capacity from performance
FR4	Upgrade and recovery shop items	To provide meaningful purchase choices
FR5	Time-limited shop countdown after standard stages	To operationalise scarcity cues through time pressure
FR6	HUD and shop interface displaying key player information	To support pressured decision-making
FR7	Stable combat, collision, death, revive, and stage transition logic	To maintain repeated playable sessions

Table 3.1 Requirement summary

The first requirement was direct player control. The game needed responsive aeroplane movement and constant firing so that the prototype could function as an action-based survival shooter rather than a passive task. The second criterion was to maintain battle pressure through enemy generation, stage scheduling, and boss encounters. This was required since purchasing decisions only make sense when players anticipate ongoing and rising difficulties.

The third criteria were a clearly defined in-game economy. The prototype needed to reward combat and survival while maintaining scarcity in accessible purchase resources. This requirement was met by segregating scrap from score and tying store access to stage progression. A major system upgrade was the fourth requirement. The prototype had fire speed improvements, damage upgrades, shield protection, and health recovery, with certain options growing more costly with additional purchases, because shop items had to offer real trade-offs. Providing an operational and apparent scarcity cue was the last requirement. This is the main prerequisite for the project. The first shop does not feature a 15-second countdown timer, but subsequent shop visits do. Time pressure becomes a real gameplay element when the timer automatically closes the purchase window. A supporting interface that could simultaneously display time, health, currency, score, shield condition, and upgrade level was the sixth need. This was required since the effect of the scarcity cue relied on the simultaneous visibility of risk, resources, and reward in addition to the timer itself.

System stability was the final requirement. Core interactions that must be constant over numerous play sessions include combat collisions, damage processing, death, revive logic, shop opening and closing, and level transitions. This requirement influenced the code's modular structure as well as the event-based communication between the user interface and gaming components.

When considered as a whole, these parameters distinguish the prototype as a full-featured timed-store survival shooter rather than a simple shop simulation. Furthermore, they serve as the framework for the following sections, which delve into greater detail into the architecture, gameplay processes, and scarcity-based shop design.

3.3 System Architecture

The prototype's modular scene-based architecture separates system entry, game execution, interface display, and shared configuration. This structure was designed to keep the prototype manageable while also allowing for testing, balance, and eventual improvement. The system distributes the work across distinct modules rather than putting all the logic in a single implementation layer.

MenuScene provides the initial UI and difficulty selection, while PreloadScene loads graphic assets to configure the runtime environment. These two scenes establish the stage for the gaming session before it begins. Because the prototype requires a smooth transition from asset preparation to player-controlled gaming, their role is limited but necessary.

GameScene is responsible for the core runtime logic. This scene manages everything from player spawning to enemy formation, projectile interaction, combat collisions, stage time, score and scrap updates, boss activation, and resurrection logic. To put it another way, GameScene manages the prototype's operational state while it is being played, as well as the rate of battle, awards, and level progression.

The prototype uses a separate UIScene to avoid overwhelming the gameplay layer with interface logic. This scene manages health displays, score displays, scrap displays, upgrade statuses, countdown timers, shop overlays, and revive prompts. Because the project relies on both visible purchasing pressure and actual combat, it is critical to separate GameScene and UIScene. By segregating these tasks, the system can maintain a clearer structure while keeping battle and interface states synchronised.

The prototype employs entity-specific classes for the player, enemies, and projectiles, while the configuration layer centralises numerical rules. The Player class handles movement, fire, damage, shield use, and immunity. The Enemy class defines threat behaviour, reward logic, and boss-specific activities. While config.js contains common settings such as health, time, prizes, and upgrade parameters, the Projectile class manages missile behaviour for both player and opponent attacks. Because key values may be modified without affecting the entire scene structure, this strategy reduces redundancy and improves system balance.

Updates prompted by events are the major form of communication inside architecture. GameScene generates events whenever major gameplay states occur, such as health loss, score increases, level transitions, shop activations, and revival requests. When these events occur, UIScene reacts by modifying the displayed interface. For a prototype in which fighting systems, interface systems, and time-limited shop behaviour must coexist without becoming structurally unstable, this approach reduces the direct coupling between gameplay and display logic.

When everything is considered, the architecture provides the project with a suitable technical foundation. Because the time-limited shop mechanism hinges on the synchronisation of combat, advancement, economics, and interface components, a modular and event-driven structure is needed to support the prototype as a unified design artefact rather than as a collection of separate aspects.

3.4 Player Control and Combat Design

The player control and fighting mechanism was designed to produce continuous action, readable feedback, and a high level of survival pressure. Since the project revolves around time-limited purchases made during gaming, combat must provide enough danger and resource demand to justify upgrades and recovery supplies.

Player movement is based on direct keyboard input, allowing for unfettered directional movement during combat. At the same time, the aircraft can shoot constantly while the firing key is depressed. This control scheme was chosen because it is basic, responsive, and appropriate for a survival shooter whose location and repeated attack movements must be straightforward to understand.

The defensive system combines temporary immunity, shielding, and health. Heart icons reflect the player's initial health poll. Direct damage initiates a brief immunity phase with a flickering effect, whereas a paid shield absorbs one hit before disappearing. These solutions reduce unfair repeated hits and highlight damage statuses. Figure 3.3 displays a typical gaming interface used in a battle stage, demonstrating how this control and fighting capabilities appear during active play.

Figure 3.3 Combat interface during a standard gameplay stage

Figure 3.3 shows that the player receives battle and resource information simultaneously, allowing for swift survival decisions during active gaming.

Opponent pressure is generated by frequent projectile threats, direct impact risk, and increasingly powerful opponent kinds. The fighting model's structure is purposely simplistic, as the prototype's primary goal is to investigate scarcity-based purchasing pressure rather than advanced combat intricacy.

A key feature of the combat system is the direct effect of shop upgrades on later performance. The speed upgrade reduces the interval between player shots, while the power upgrade increases projectile damage. The effect of the two main offensive upgrades can be expressed in simplified form as follows:

$$R_n = \frac{R_0}{1 + 0.1n}$$

$$D_n = D_0(1 + 0.1n)$$

where R_0 and D_0 represent the base firing interval and base projectile damage, and $n$represents the upgrade tier.

To summarise the key combat-related player attributes and their gameplay role, Table 3.2 lists the main player combat properties and upgrade effects.

Attribute	Base state	Upgrade effect	Gameplay purpose
Movement	Free directional movement	No tiered upgrade	Supports dodging and positioning
Firing interval	Base weapon rate	Reduced by speed tier scaling	Increases offensive output
Projectile damage	Base missile damage	Increased by power tier scaling	Improves elimination efficiency
Health	Fixed starting health	Can be restored through shop purchases	Supports survivability
Shield	Inactive by default	Absorbs one hit when purchased	Adds defensive protection
Immunity	Triggered after damage	Temporary invulnerability	Prevents repeated immediate damage

Table 3.2 Player combat attributes and upgrade effects

These traits collectively characterise the combat pressure, which has practical implications for following shop decisions. The shop's scarcity cue would have no effect on the gaming loop if there were no repeated damage risk, upgrade demand, or survival strain.

3.5 Enemy, Wave, and Boss Design

The enemy and progression mechanism aimed to increase the level of pressure as the stages advanced. Because the project bases purchase decisions on scarcity cues, the gaming loop required a threat structure that could provide relevance to later upgrades and recovery decisions. Rather than having a single recurrent enemy pattern, the prototype employs a diversity of enemy types, progressive spawning pressure, and sporadic boss battles.

The prototype has four basic enemy types: scout, ordinary, elite, and boss adversaries. These groupings differ in terms of attack strategy, reward value, scale, movement speed, and health. Typical adversaries present the standard level of battling challenge, whereas scout enemies are lower-level threats with limited resilience. Boss enemies are major stage checkpoints, with higher health, a larger visual scale, and more powerful attack behaviour, whilst elite foes are more powerful and yield higher scrap rewards. While retaining a clear battling framework, this tiered method increases the variety of foes.

Enemy pressure is also shaped by stage rhythm. During standard stages, enemies spawn repeatedly over a fixed combat period, while the spawn interval decreases over time according to a difficulty ramp. This means that even within a single run, the player faces denser pressure as the session continues. In addition, the proportion of stronger enemy types increases over time, which raises the cost of entering later stages without sufficient upgrades or recovery. This progression logic is important because the scarcity cue in the shop depends on anticipated future threat rather than on the current stage alone.

To offer distinct advancement peaks, boss encounters are introduced at predefined stage intervals. Every five stages in the prototype, a boss stage substitutes the regular battle transition with a lengthier combat encounter. There are three ways that boss enemy vary from normal enemies. First, their health has greatly improved. Second, they use their distinctive projectile habit and attack method. Third, the gameplay cycle changes significantly when they are defeated. This rhythm heightens the sense of progress during the session and breaks up the monotony of typical waves.

The spawn interval function is a simple technique to illustrate increasing enemy pressure. If S_0 represents the initial spawn interval, then the interval after k difficulty updates can be represented as:

$$S_k = max(S_{min}, S_0 \cdot \gamma^k)$$

where $S_{\min}$ is the minimum allowed spawn interval and γ is the difficulty ramp factor, with $\gamma < 1$. In the implemented prototype, the spawn interval decreases gradually but is bounded by a lower limit so that enemy generation remains challenging without becoming unmanageable.

To summarise the major enemy categories and their gameplay role, Table 3.3 presents the main enemy types, their combat characteristics, and their function in the progression system.

Enemy type	Main characteristics	Reward role	Progression role
Scout	Low health, fast movement, light threat	Low scrap reward	Entry-level pressure
Standard	Moderate health and baseline combat behaviour	Standard scrap reward	Core wave threat
Elite	Higher health, stronger presence, projectile threat	Higher scrap reward	Increased stage pressure
Boss	Very high health, larger scale, stronger attacks	Major reward and stage transition value	Progression checkpoint

Table 3.3 Enemy types and progression roles

The enemy system combines escalation and readability, as seen in Table 3.3. The boss stage is a significant progression marker rather than a simple statistical fluctuation, and each enemy category adds a new level of difficulty.

These design decisions boost the challenge of the fight cycle while also making the store system functional. In the prototype, following purchase decisions would be less urgent and strategically important in the absence of increased enemy pressure, elite threats, and boss encounters.

3.6 Upgrade and Economy System Design

The upgrading and economy mechanism was designed to convert battle effectiveness into strategic decisions later. Because the project is based on scarcity cues in a time-limited shop, the prototype required a resource system capable of producing major trade-offs rather than automatic advancement. As a result, the game distinguishes between score and scrap, and store selection is closely related to combat survival.

Scrap is used as a resource for in-game purchases. It is collected by defeating foes and completing stages, and it may be used to buy upgrades and recovery supplies from the store. In contrast, score measures gameplay performance and advancement rather than purchasing power.

This distinction is important because it prevents the shop system from becoming reduced to a simple representation of score accumulation. Even with a high score, a player may be forced to make difficult judgements about what to buy if scrap remains limited.

The shop includes four major purchase categories related to the main game loop: shield protection, health recovery, projectile power upgrades, and firing speed increases. The game also incorporates a respawn mechanism as part of its bigger survival economy concept. These options were chosen because they have very different consequences for succeeding in combat. Some improve defensive stability, while others increase offensive output and promote error recovery. As a result, the player must allocate scarce scraps among competing short- and medium-term needs.

The upgrade system also includes cost progression. Speed and power upgrades do not remain fixed in price across repeated purchases. Instead, their cost rises with tier level, which prevents unrestricted early accumulation and preserves scarcity in available resources. This design makes the purchase process more strategic because the player must decide not only what to buy, but also whether to invest in repeated offensive upgrades or to preserve scrap for health, shield, or later recovery needs. In simplified form, the cost of an upgrade at tier $n$can be represented as:

$$C_n = C_0 + n\Delta C$$

where C_0 is the base cost of the upgrade and ΔC is the cost increment applied at each successive tier. This linear cost structure reflects the implemented pricing logic for repeated offensive upgrades in the prototype.

The relation between combat rewards and later purchasing choice can also be described in simplified form. Let total available scrap after a stage be represented as:

$$M = M_0 + R_k + R_s - \sum C_i$$

where M_0 is the scrap carried into the stage, R_k is scrap gained from enemy kills, R_s is stage survival reward, and $\sum C_i$ is the total cost of purchased items. This expression is not used as a runtime formula in the code, but it clearly represents the economic logic of the prototype: combat success expands purchasing options, while previous purchases reduce later flexibility.

To summarise the major purchasable items and their design role, Table 3.4 lists the main shop options, their economic behaviour, and their function in the survival loop.

Shop item	Economic behaviour	Main gameplay effect	Decision role
Speed upgrade	Cost increases with repeated purchase	Raises firing rate	Offensive investment
Power upgrade	Cost increases with repeated purchase	Raises projectile damage	Offensive investment
Shield	Fixed-cost purchase	Absorb one hit	Defensive protection
Health restore	Fixed-cost purchase	Restores health	Short-term survival recovery
Revive	Cost increases with repeated use	Restores play after death	Emergency continuation option

Table 3.4 Shop items and economic role

The shop system is not dependent on duplicate alternatives or cosmetic variety, as seen in Table 3.4. Time-limited shop decisions are more important because each purchase category has a distinct effect on following combat.

This design helps the project achieve its ultimate purpose in two ways. First, by limiting resources and distinguishing item functions, it transforms the shop into a true decision-making environment. Second, because the player must make time-sensitive purchasing selections while under resource constraints and anticipating future battle demands, the scarcity cue has genuine meaning.

3.7 Time-limited Shop Pressure Mechanism

The key design element of this project is the time-limited shop system. Rather than employing a static store interface, it intends to turn shortage indicators into a fun gaming environment. Due to a limited purchasing window that opens between battle stages, the prototype forces the player to make decisions about upgrades and recoveries within a fixed amount of time. This approach incorporates purchase pressure inside the gameplay loop rather than placing it outside of it.

A temporal framework based on stages incorporates the shop system. Before engaging in regular combat, the player enters an initial store at the beginning of a session. Because there is no time constraint at this first store, the player is free to choose their opening buy. The game then switches between 30-second fighting and shop sessions. The player is moved to a store interface at the conclusion of a normal level, where purchases must be finished in 15 seconds before the subsequent phase automatically starts. To avoid the scarcity queue being present at every purchase moment in the same way, a clear distinction was developed between an untimed initial transaction and subsequent time-limited shops. Rather, the player initially encounters the business in a low-pressure scenario, after which it is expressly under time constraint.

The store system's temporal logic can be summarised as follows.

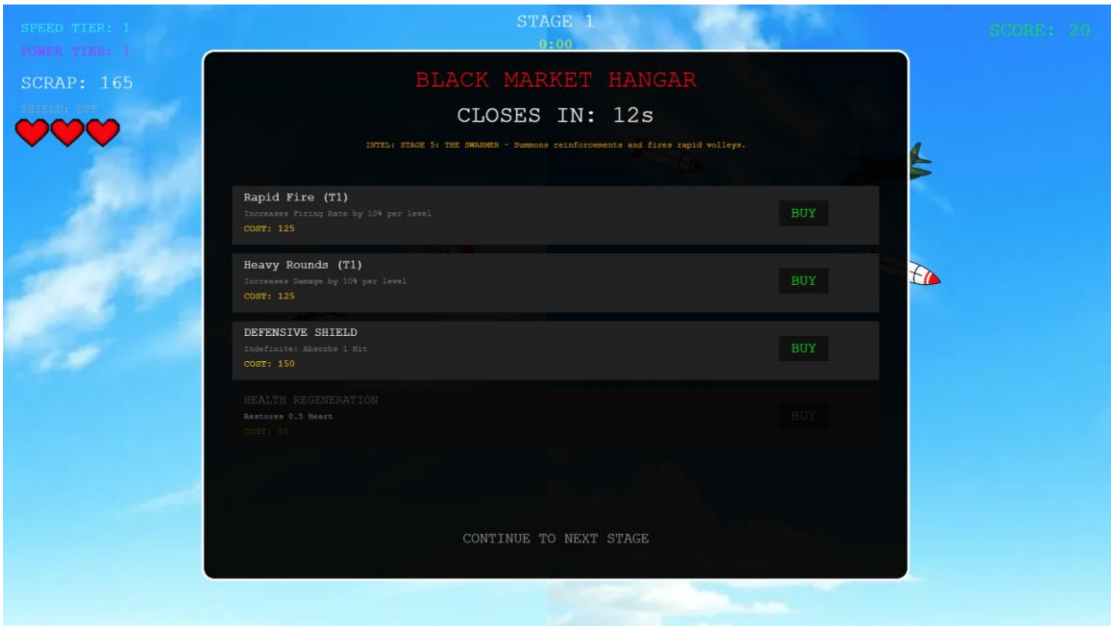
$$T_{stage} = 30\text{s}, T_{shop} = \begin{cases} \infty, & \text{for the initial shop} \\ 15\text{s}, & \text{for subsequent shops} \end{cases}$$

where T_{stage} denotes the duration of a standard combat stage and T_{shop} denotes the duration of the shop phase. This expression formalises the central timing rule of the prototype and makes clear that the initial and later purchase situations are intentionally differentiated.

The timer's behavioural function goes beyond just counting down the seconds. The timing alters the decision-making process. In normal shopping circumstances, the player could examine products with minimal immediate prices for hesitation. In a time-limited shop, the purchasing window automatically closes, making procrastination pricey. The player must balance several elements at the same time, including the current scrap, health state, shield status, offensive upgrade level, and projected difficulty of the next stage. This design makes time an active component of the decision environment, while also compressing discussion.

Figure 3.4 compares the later time-limited shop interface used during stage progression to the first untime-limited shop to demonstrate the visual differences between the two purchasing scenarios.

(a) Initial store, No Countdown



(b) In-game store, 15-second countdown

Figure 3.4 Comparison of the initial shop and time-limited shop interfaces

As shown in Figure 3.4, the later shop phase introduces a visible countdown that changes the player's decision context from open selection to time-limited allocation. This visual difference is important because the scarcity cue in the prototype is communicated not only through the underlying timing rule, but also through interface presentation.

Automatic state transition is also necessary for the time-limited shop's architecture. When the timer runs out, the shop closes and the next gaming phase begins without the player's agreement. Scarcity signals are only meaningful when the purchasing opportunity is truly limited, which is why this criterion is necessary. The technique would not function as a true scarcity condition if the timer was purely decorative and allowed the player to continue purchasing after it expired. As a result, the prototype sees the countdown not merely as a visual effect, but also as an actual

system limitation.
This mechanism supports the overall aim of the project in two ways. First, it provides a clear implementation of scarcity cues inside a game system. Second, it creates a purchase environment where urgency, limited resources, and anticipated future threat interact directly. The time-limited shop is therefore not an isolated interface feature. It is the point at which combat results, economic constraints, and progression pressure converge into a single decision moment.

3.8 UI and Feedback Design

The prototype's user interface was designed to enable quick reading, ongoing state awareness, and swift decision-making. Because the research examines purchase intention in the context of scarcity cues, the interface needed to do more than just display basic statistics. To allow the player to make quick decisions during the battle and shop phases, risk, resources, and progression status needed to be clearly displayed.
The UI displays the current stage, remaining stage time, score, scrap, offensive upgrade levels, shield state, and heart-based health while the game is in action. The simultaneous visibility of these components enables the player to monitor both short-term survival and long-term advancement without ever leaving the combat screen. Figure 3.3 shows that combat and resource information are provided at the same time to allow for quick decision-making during active gameplay.
Additionally, the interface directly supports the scarcity mechanism. To keep the short purchasing window visible during the decision-making process, the countdown timer is prominently displayed in succeeding store phases. This shift is visible in Figure 3.4, as the player moves from an untimed initial shop to a later store interface with an explicit time limit. In this way, both the interface display and the underlying temporal rule communicate the scarcity cue.
Defensive feedback was designed to improve readability rather than overly detailed displays. Heart icons represent health, shield state is displayed separately, and when a player is defeated, revival prompts are displayed via a specific overlay. These traits were chosen since the prototype relies on quick interpretation both during timed window purchases and during combat. As a result, rather than acting as a cosmetic overlay, the interface is a part of the gaming system.
Overall, by providing survival, economy, and timing data in a single decision context, the UI design helps to achieve the project's goal. The timer, player state, and upgrade status would be more difficult to grasp without a clear interface, reducing the time-limited shop's intended buy pressure.

3.9 Testing and Refinement Process

Rather than a single linear implementation pass, the prototype was refined incrementally and through functional testing. This technique was acceptable because the project includes numerous interacting systems, including combat, enemy spawning, shop timing, upgrade scaling, revival logic, and interface feedback. As a result, rather than depending solely on isolated implementation, prototype stability depended on the synchronisation of many systems.
The first phase of testing focused on core functions. This featured player movement, continuous fire, enemy spawning, projectile behaviour, collision management, damage processing, and stage time. The key goal at this time was to ensure that the prototype could maintain a full battle loop without interfering with basic gameplay interaction. Later testing concentrated on the

transitions between fight and shop phases, countdown behaviour, boss-stage activation, reviving prompts, and automatic shop closure. Because the shop's dependability within the stage framework, rather than as a standalone interface component, is critical to the project's underlying mechanic, these assessments were extremely important.

Balancing was the focus of a second refinement phase. The time-limited shop's significance throughout play is influenced by enemy pressure, spawn timing, reward values, upgrade prices, and revival costs. When resources build up too quickly, there is less trade-off when making purchases, which weakens the scarcity queue. The store might feel punitive rather than coercive if the level of difficulty increases too quickly. Therefore, refinement entailed changing system settings so that the player frequently had to make difficult but doable choices.

The final phase of enhancement focused primarily on interface readability. Because the research investigates scarcity cues using a time-limited shop, the user had to quickly detect the countdown and analyse the available resources. As a result, during live play, testing considered the visibility of health, scrap, score, upgrade levels, and timing information. When these testing iterations were integrated, they helped the prototype become a unified-timed-shop survival system that emphasised both functional stability and design brilliance.

Overall, the refining process contributed to the creation of a stable design artefact from a collection of disparate systems. The final prototype does not reflect a finished commercial game. However, it does create a playable setting in which scarcity cues, battle pressure, and upgrade options may be examined concurrently inside a structured gaming cycle.

4. Results / Findings

4.1 Final Implemented Game Features

The final prototype was a playable browser-based 2D survival shooter with combat, progression, resource management, and a time-limited shop system all in one gameplay loop. The finished system includes player-controlled aircraft movement, constant missile fire, enemy wave creation, level progression, monster encounters, upgrade purchasing, health and shield management, score tracking, scrap collection, and revival logic. Rather than a discrete shop simulation, these pieces combine to form a working game prototype.

The combat level supports directional movement, repeating projectile fire, enemy impacts, projectile damage, and momentary immunity after player damage. The prototype features automatic transitions between fighting and shop phases, recurrent 30-second stages, and sporadic boss stages at the progression level. The concept separates score and scrap at the economic level, allowing the player to spend earned scrap on both offensive and defensive purchases. The prototype provides visible data on health, score, scrap, stage timer, upgrade level, shield state, and revive conditions at the interface level. When viewed as a whole, these realised elements show the project's successful transition from conceptual design to fully functional prototype.

Most importantly, the prototype successfully implemented the project's key scarcity mechanism. Later shop visits use a visible 15-second buy window, while the first shop does not have one. This suggests that the prototype's final form expands beyond a conceptual explanation of temporal scarcity. It connects that technique to resource-constrained upgrade options and integrates it directly into playable stage development. This is the major feature that distinguishes the prototype from a typical small-scale survival shooter in terms of project outcomes.

4.2 Gameplay Flow Across Stages

Before beginning the session, the player selects a difficulty level from the main menu, triggering the final gameplay sequence. This entry point takes you directly into the first shop phase and sets the run's initial conditions. The initial store is not timed, unlike future stores, allowing the player to make a low-pressure opening decision before the commencement of regular warfare. This design establishes the first difference between unrestricted purchasing options and future time-limited choices.

Figure 4.1 Main menu and difficulty selection

Figure 4.1 depicts the menu screen for starting a session and selecting the difficulty. Despite its simple design, this interface establishes the initial condition for the entire gaming cycle.

When the shop opens, the player enters the standard battling stage. During this phase, the player must maintain control over mobility, attack timing, and damage avoidance to defeat or survive opponent waves. Score increases as the battle progresses, but scrap is acquired by defeating adversaries and completing the stage. When the stage timer expires, the player advances to the next shop phase, where resources can be invested for improvements or recovery. The swing between buying and battling creates the prototype's main rhythm.

The game loop becomes increasingly intense as the run progresses. The player enters later stages with rising upgrade requirements and survival risk as enemy density grows, and stronger enemy types appear more frequently. Boss encounters throw off the typical stage rhythm and act as a progression peak every five stages. Boss stages differ from standard levels in that they provide a more substantial and long-term threat, increased attack pressure, and a clear milestone progression. Because of its structure, the session follows a predictable pattern of scheduled purchases, frequent combat, and sporadic high-pressure clashes.

Figure 4.2 Boss battle

Figure 4.2 depicts a boss battle scene, which represents an important progression checkpoint in the prototype. In comparison to other combat stages, the boss encounters intensify the threat and stresses the importance of early upgrade and recovery options.

The session does not end immediately if the player is defeated. Rather, the prototype provides a revitalised interface with limited opportunities for future evolution at an increasing cost. This feature broadens the game's economic logic by adding defeat as another resource-based decision point. As a result, the gameplay flow extends beyond merely combat and shopping. The prototype's overall pressure structure is reinforced by the inclusion of emergency survival decisions with limited resources.

In general, the gaming flow functions as planned. A recurring sequence of entry, fight, reward, store selection, renewed combat, and recurring boss escalation are all part of the prototype. This attests to the successful implementation of a full playable loop in which timed by pressure is incorporated into advancement as opposed to being inserted as a stand-alone feature.

4.3 Shop Interaction and Time-limited purchase Behaviour

In the final prototype, the shop interaction system created two separate purchasing scenarios. The player enters an untimed opening shop before the start of regular combat in the first context, which occurs at the beginning of a session. With no immediate time constraints, the player can explore the numerous objects in this scenario and determine whether to make an initial purchase based on strategy or personal preference. In the second context, which comes after the latter levels, the player returns to the shop with a visible 15-second countdown. The same type of purchasing decision is made in this case, with a clear time constraint. This difference, which transforms the shop from a generic upgrading interface into a scarcity-based choice environment, is one of the prototype's most significant results.

Later shop phases demand the player to disperse a limited amount of scrap while considering the character's current health, shield condition, offensive upgrade level, and expected future threat. Practically speaking, this means that the store is not regarded as a straightforward inventory. It is viewed as a small window of chance to select between two dangerous game conditions. Before the next stage begins, the player must make a quick decision on how much to spend on attack, defence, or recovery after entering the store and seeing the outcome of the previous stage. This interaction pattern is the best example of how the scarcity mechanism was successfully utilised at the gaming flow level.

Figure 3.4 already showed how the timed and untime-limited shop states differed visually. In terms of consequences, the distinction affects how the session's purchase phase operates. The untimed shop supports planning, but the time-limited shop supports limited allocation under pressure. Because the player has limited resources and little time to decide how to spend them, the latter purchase phase acts as a behavioural bottleneck in the gameplay loop.

The prototype does not claim to provide formal behavioural proof that the timer increases purchase intention across a participant sample. However, as a design outcome, it clearly establishes a playable context in which this effect may be examined. The prototype effectively operationalizes scarcity cues by implementing actual purchasing restrictions, a visible countdown display, and an automated shop closing. These design outcomes are sufficient to show that the project created a functioning setting for future behavioural research and went beyond conceptual discussion.

In general, the shop interaction mechanism works as expected. It creates a controlled contrast between timed and untimed decision settings and connects them to important gaming outcomes. This is the main design consequence of the project.

4.4 Balancing Outcomes of Upgrades and Enemies

Furthermore, the final prototype struck a reasonable mix between resource constraints, upgrade value, and opposing pressure. This does not imply that the prototype achieved an optimal commercial balance. It does, however, show how the mechanisms that have been implemented interact in a way that makes shop decisions important when playing. Stronger adversaries appear more frequently in succeeding rounds, the threat posed by attackers grows over time, and boss encounters results in unique pressure peaks. The player can simultaneously boost their battle performance by employing a shield, improving their speed, power, and health. As a result, the value of these acquisitions is determined by the player's remaining resources and current growth rate.

One useful effect of the balancing design in terms of performance is that offensive and defensive purchases remain functionally separate. Speed and power boosts increase damage output, while shield and health recovery improve survival. This means that when it comes to making shopping decisions, there isn't simply one apparent ideal option. Rather, the player must react to the current state of play. While a player with higher survivability may invest in offensive efficiency to prepare for future pressure, a player with poor health may prioritise recovery. The time-limited shop has meaningful design values only when multiple purchasing options remain highly competitive, making this type of trade-off critical.

The economic system also demonstrates the connection between balance and scarcity. Scrap is scarce enough that the player cannot buy everything at will, yet it is still available frequently enough to keep the shop relevant at the end of each stage. As a result, the purchasing environment is limited yet nevertheless functional. A modest scrap collection would render the shop obsolete. The timer would be less significant if it were too high because the player might purchase without having to make a significant trade-off. The current implementation maintains the shop under pressure to advance while avoiding both extremes.

Boss encounters also reinforce this equilibrium system by serving as milestones for advancement. Their existence adds value to previous upgrade and recovery decisions while increasing the perceived cost of poor preparedness. Thus, stronger rivals are not the exclusive cause of later-stage equilibrium. It is the result of a combination of a time-limited purchase window, money restrictions, and enemy escalation. That interaction is one of the most important outcomes of the completed prototype since it provides the scarcity cue with useful meaning within the game system.

When combined, these findings show that the prototype created a workable balance framework that was adequate for the project's goals. While preserving a playable loop in which the player can advance, modify, and recover, the linked systems create enough pressure to make significant decisions. Even more intensive playtesting and adjustment would be required for a more precise final balancing model, but this is an acceptable output for a capstone design prototype.

4.5 Observed Strengths and Issues

The final prototype has several obvious design advantages. First, it seamlessly integrates economy, time-limited purchases, progression, and combat into a single entertaining cycle. An isolated store screen is not linked to the timer. Rather, it fits within the larger context of expected future threat, resource gain, and stage completion. Instead of simply using the scarcity cue as a visual gimmick, it now has functional significance within the game system.

Another feature of the prototype is the transparency of its basic systems. Combat feedback is quick, player control remains intuitive, and shop economics is simplified by distinguishing between score and scrap. The distinction between untimed and time-limited shop phases is also effective since it offers the prototype with a clear internal contrast between lower-pressure and higher-pressure purchasing environments. As a result, the study provides a useful framework for future research into time-limited purchase pressure.

The prototype is similarly limited in other respects. The most notable aspect is that the experiment does not include official participant testing or substantial behavioural data. As a result, the study can discuss how the time-limited shop may influence players' intentions to make purchases, but it cannot claim to have experimentally proven that effect across all participants. A second disadvantage of the balancing model is that it remains at the prototype level rather than production level. Although reward pacing, upgrade tweaking, and adversary behaviour are all functional, additional playtesting and iteration may be desirable.

Another issue is that the prototype values system clarity over content diversity. This is appropriate for the project's purpose, but it also implies that the current version has a small item pool, a single major play style, and a limited enemy set. These constraints limit the number of behaviours that can be observed in a single session, but they do not jeopardise the project's design rationale. Overall, the data show that the prototype works well as a design artefact while still leaving plenty of space for future refinement and growth.

5. Discussion

5.1 Interpretation of Gameplay Outcomes

The project's results show that a stage-based survival shooter may successfully add a timed scarcity component. The shop is not regarded as a distinct menu or as a purely cosmetic feature in the final version. Rather, the player must make decisions on purchases in between battle rounds depending on their current performance, available resources, and predicted future pressure. This is a noteworthy finding since it shows that scarcity cues may be used in system design as well as interface text and advertising layouts.

The findings also show that the time-limited shop performs best when the surrounding gameplay structure supports it. This project makes use of more than just a countdown. The player's limited scrap, evident health risk, major upgrade options, and increasing enemy pressure all influence how it functions. The timer would be much lighter without such supplementary components. This is significant because it shifts the conversation's focus from "does a timer exist" to "what design conditions make a timer matter."

Another way to look at it is that the initiative supports approaching monetization methods such as gameplay structures rather than external commercial tools. The prototype replicates the fundamental logic of scheduled purchase pressure, albeit using earned in-game currency rather than real-money transactions. This avoids the immediate financial risk of commercial adoption and makes the project useful for testing scarcity cues in a more controlled design environment.

5.2 Influence of Time Pressure on Purchase Behaviour

The findings of this project are consistent with the literature reviewed in Chapter 2 in two main ways. First, they lend credence to the notion that impatience and a lack of thought give scarcity cause more strength. The prototype's time-limited shop creates a confined window for making decisions, which is consistent with previous studies indicating that time limits can encourage people to make decisions more swiftly and emotionally. The project implements the same design thinking in a playable gaming environment but does not statistically test the effect.

Second, the findings back up the hypothesis that the larger service or gaming system, rather than just the object, drives virtual buying activity. The value of an upgrade in this project is determined by stage progression, available scrap, enemy pressure, and current health. This is consistent with previous studies revealing that the system in which virtual products are employed provides significance. Thus, the prototype supports the idea that interface design alone is insufficient to understand time-limited purchase behaviour in games. Furthermore, it must be understood through progression design, risk structure, and economic design.

This project also examines the literature on dark patterns and interface-based urgency. The prototype shows that even without real-money payment, a visible countdown, automatic closure, and shortened decision window are enough to create a strong scarcity situation. This adds support to the theory that behavioural pressure in digital systems is usually caused by both the interface design and the surrounding decision environment. However, as compared to direct commercial implementation, the prototype provides a cleaner setting for examining this mechanism because it employs earned currency and clear time.

5.3 Strengths of the Current Design

The project's obvious connection of theory and practice is one of its strong points. The report starts with literature on temporal pressure, virtual commodities, and scarcity before translating those concepts into a functional prototype with a particular gameplay structure. Compared to a prototype that only has a timer and no apparent conceptual aim, this gives the project more cohesion.

The capacity to combine many systems into a single loop is a second strength. The interplay between the fighting system, level advancement, enemy escalation, scrap economy, time-limited shop, and revival logic provide support for the major study topic. This integration is critical because the project involves more than just demonstrating discrete features. It demonstrates how such factors combine to create pressure when there is active play.

A third strength is the relative clarity of the prototype. The system is nonetheless extensive enough to provide major trade-offs between assault, defence, and recovery, while being modest enough to be simply comprehended. This balance of scope and coherence is essential for a capstone design project. The prototype is large enough to be thoroughly inspected without diminishing the basic concept.

5.4 Limitations of the Project

The main limitation of the project is the lack of rigorous user testing. Although the prototype provides a believable context for researching timed scarcity, the paper cannot conclude that the timer significantly enhanced purchase intention for players as a quantifiable behavioural result. Currently, the project promotes design analysis rather than causal population-level inference.

A second constraint is that the prototype's content scope is still simplified. Enemy variation, store item diversity, and gameplay options are purposely limited to make the scarcity mechanism easier to isolate and evaluate. This improves methodological clarity, but it also lowers the behavioural variation that could occur in a larger or more content-rich game.

A third restriction is maturity in balance. Although the prototype shows potential links between awards, upgrades, and opponent escalation, these systems have not been thoroughly tested or developed over time. As a result, the balance results shown in this report should not be considered final design optimisation, but rather prototype-level results.

5.5 Implications for Small-Scale Game Design

According to this project, timed scarcity in games should be viewed as a system problem rather than a simple interface decision. Only when a countdown is supported by limited resources, uncertainty about the future, and distinct purchasing options does it become meaningful. Designers must consider the decision pressure that surrounds a timer in addition to its display.

The project also shows how earned in-game currency may be utilised to investigate scarcity cues in non-commercial prototypes. This is useful for design research since it allows for the analysis of monetization-style pressure without putting real money at risk. This provides a useful balance between practical commercial testing and abstract theory for academic study.

The findings also show the importance of being prudent while developing monetization tactics. A timer can become immoral if it is used in conjunction with direct payment, opaque value structures, or deceitful framing. This project shows how the underlying logic can be built and studied, but it does not apply that strong form of pressure.

5.6 Future Work

Formal player testing is the most obvious next step. A subsequent study might examine purchase frequency, decision speed, item choice, or stage survival under both timed and untime-limited shop conditions using a sample of players. This would enable empirical testing of the design reasoning presented in the current project.

Wider content variation is a second avenue for future research. To investigate how scarcity cues interact with greater gameplay diversity, more enemy kinds, wider item categories, more intricate boss patterns, or different shop layouts might be implemented. Examining time-limited purchase pressure under a broader range of strategic circumstances would be conceivable as a result.

Comparing ethics is a third direction. To investigate the boundary between persuasive and manipulative design, future research may compare transparent clocks, less noticeable timers, and more forceful framing. This would enable the project to contribute to discussions on ethical game monetization techniques as well as gameplay design studies.

6. Conclusion and Recommendations

This project created and built a browser-based 2D survival shooter with scarcity cues integrated into a time-limited shop. Combat, stage progression, resource management, and purchasing options are all integrated into a continuous gameplay loop in the prototype. Its major design aspect is the contrast between an untimed opening shop and later shop stages, which are limited by a visible 15-second countdown. This framework was used to investigate how time-limited purchase pressure can be implemented through active gaming rather than a static commercial interface.

The final prototype successfully included the desired scarcity mechanism as a functional component of the game system. The findings demonstrated that time-limited purchase pressure can be integrated into a survival shooter cycle in which enemy escalation, limited scrap, visible player state, and upgrade option all influence decision-making. The experiment also demonstrated that the effect of the timing is dependent on the surrounding gameplay structure. The countdown gains significance because it is tied to combat risk, resource constraints, and expected future threats.

This report does not claim to provide formal behavioural proof that the time-limited shop increases purchase intention across a player population. It does, however, offer a functional design artefact that converts concepts from literature into a playable prototype. The project contributes in two ways. It first shows how to operate scarcity cues in a gaming environment. Secondly, it offers an organised prototype that can facilitate further empirical and design-focused study.

Overall, the project shows that game design can be an effective method for researching timed scarcity. The prototype takes the conversation beyond abstract theory and provides a solid foundation for future study on player decision-making under time limitations by including purchase pressure into a comprehensive gaming loop.

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Appendices

Appendix A. Parameter Tables

This appendix summarises the main numerical parameters used in the final prototype. These values support the balancing structure described in Chapter 3 and provide supplementary detail on stage timing, combat behaviour, rewards, and shop-related progression. Detailed parameter reporting is included here to keep the main chapters concise while preserving transparency in the design configuration.

Parameter	**Value**	**Notes**
Screen width	1920 px	Main game resolution
Screen height	1080 px	Main game resolution
Player base movement speed	500	Implemented in Player.update()
Initial player health	6 HP	Equivalent to 3 full heart bars
Initial scrap	200	Starting in-game currency
Standard stage duration	30 s	Defined by SHOP_INTERVAL = 30000 ms
Time-limited shop duration	15 s	Defined by SHOP_DURATION = 15000 ms
Initial enemy spawn interval	1800 ms	Base spawn interval
Difficulty ramp factor	0.97	Applied repeatedly during play
Minimum spawn interval	600 ms	Lower bound for spawn reduction
Elite spawn chance (start)	0.05	Initial elite probability
Elite spawn chance (max)	0.30	Upper bound for elite probability
Boss stage interval	Every 5 stages	Boss appears at fixed progression intervals
Survival reward	50 scrap	Granted after clearing a stage
Maximum revives	2	Hard limit on revive attempts

Table A1. Core gameplay and progression parameters used in the final prototype.

Entity / Projectile	**Health / Damage**	**Speed / Multiplier**	**Fire rate**	**Reward / Score role**	**Notes**
Scout enemy	1 HP	1.3× base enemy speed	3000 ms	5 scrap / 1 score	Entry-level threat
Standard enemy	3 HP	1.1× base enemy speed	2500 ms	10 scrap / 2 score	Baseline wave enemy
Elite enemy	7 HP	0.8× base enemy speed	2000 ms	25 scrap / 4 score	Stronger projectile threat

Boss enemy	100 HP + 20 × stage	0.3× base enemy speed	2000 ms	300 scrap / 10 score	Major progression checkpoint
Player standard projectile	1 base damage	1000	450 ms base fire interval	—	Damage and fire rate scale with upgrades
Enemy projectile	1 damage	550	—	—	Standard enemy shot
Boss projectile	2 damage	550	—	—	Higher-damage enemy shot

Table A2. Enemy and projectile parameters used in the final prototype.

Item / Upgrade	**Base cost**	**Cost growth**	**Main effect**	**Additional notes**
Speed upgrade (Rapid Fire)	100 scrap	+25 per tier	Increases firing rate by 10% per level	Maximum tier = 10
Power upgrade (Heavy Rounds)	100 scrap	+25 per tier	Increases projectile damage by 10% per level	Maximum tier = 10
Defensive shield	150 scrap	Fixed	Absorbs 1 hit	Indefinite until hit
Health regeneration	50 scrap minimum	None	Restores 0.5 heart	Cost varies with current health state
Revive	300 scrap base	+150 per use	Restores play after death	Limited to 2 revives

Table A3. Shop items and upgrade-related parameters used in the final prototype.

Appendix B. Project Management and Feedback Evidence

This appendix summarises the project management structure and the main forms of guidance that influenced the final prototype and report. The material presented here is adapted from the earlier project proposal and from the final development direction of the capstone. It is included to document how the project progressed from initial planning to final implementation and how the final report structure was aligned with the guidance of the subject and supervisor.

Stage and time	**Planned focus**	**Final outcome**
Stage 1 11/2025-01/2026	Finalise project concept, define core mechanics, and prepare technical specification	The project focus was defined as a time-limited shop 2D survival shooter examining scarcity cues in gameplay
Stage 2 01/2026-03/2026	Develop core gameplay loop and shop system	A playable browser-based prototype was implemented, including combat, stage progression, scrap economy, and shop interaction
Stage 3 03/2026-04/2026	Conduct testing, debugging, and interface refinement	Functional testing and balancing adjustments were carried out across combat, timing, economy, and UI systems

Stage 4 04/2026-05/2026	Complete final prototype and prepare design documentation	The final report was structured around Introduction, Literature Review, Method, Results, Discussion, and Conclusion

Table B1. Summary of planned project stages and final outcomes.

Stage of project	**Supervisor guidance / discussion point**	**Response in the final project**
Project initiation	The supervisor introduced the overall project direction and indicated that the project could be developed in the form of a game, with the report focusing on the game design and development process.	The project was framed as a capstone design prototype, and the final report was organised around the design, implementation, and analysis of the game rather than as a full experimental study.
Early concept development	A basic game idea involving purchasing and upgrading during progression was suggested as a suitable direction.	The project adopted a stage-based survival shooter structure with combat, resource accumulation, and in-game purchasing as the core gameplay loop.
Design refinement	To make purchasing decisions more meaningful, the gameplay needed enough difficulty and pressure to encourage players to consider upgrades.	The combat loop was designed with enemy escalation, stage progression, boss encounters, and survivability pressure so that upgrades and recovery choices would carry practical value.
Prototype review	After the initial demo was completed, feedback indicated that the opening shop should not include a countdown timer.	The initial shop was revised into an untime-limited purchase phase, creating a contrast between the first low-pressure purchase condition and later time-limited purchase conditions.
Prototype review	Feedback also suggested adding a harder gameplay mode.	A hard mode was implemented in the final prototype through a more demanding starting condition and faster enemy pressure.
Final refinement	The revised prototype was reported back after the requested changes were completed.	The final version of the game incorporated the key modifications discussed during supervision and formed the basis of the report.

Table B2. Summary of project guidance and response

Some Teams chat and email screenshot:

Haoyu Liu 25/07/2025 5:52 pm

The research questions look good. I will schedule a meeting with you next week, and I will let you know my available time slots later this week.

25/07/2025 5:54 pm

Got it. Thank you.

Sunday, 27 July 2025

Haoyu Liu 27/07/2025 9:44 pm

I am available
July 30 (Wednesday) 10:00-11:00
July 31 (Thursday) 10:00-12:00
August 1 (Friday) 10:00-12:00

27/07/2025 9:48 pm

I would like to select Thursday. I am available during that time period.

Haoyu Liu 27/07/2025 10:16 pm

Time: Jul 31, 2025 10:00 AM Canberra, Melbourne, Sydney

Join from PC, Mac, Linux, iOS or Android:
https://zoom.uts.edu.au/j/88045330086

27/07/2025 10:16 pm

Got it.

Haoyu Liu 22/09/2025 9:10 am Edited

Please schedule a meeting with me to discuss your Literature Review. I will be available
September 22 (Monday) 15:00-16:00
September 24 (Wednesday) 10:00-12:00
September 26 (Friday) 11:00-13:00
September 29 (Monday) 15:00-16:00

22/09/2025 2:29 pm

I'm available today, we can start on 15:00.

22/09/2025 3:30 pm

Or we could meet on Wednesday at 11 a.m.

Haoyu Liu 22/09/2025 4:03 pm

Time: Sep 24, 2025 11:00 AM Canberra, Melbourne, Sydney

Join from PC, Mac, Linux, iOS or Android:
https://zoom.uts.edu.au/j/84572460610

22/09/2025 4:05 pm

Got it

Saturday, 18 October 2025

Haoyu Liu 18/10/2025 10:49 pm

1. The Introduction section should present the background and motivation behind your research question. Explain why the question is meaningful, relevant and important. You may support your argument by citing news articles or industry reports.
2. Based on your Low-Risk Application Form, you have chosen to design and develop a simple game. Therefore, your proposal should focus on this process. No participant recruitment or data collection should be involved. Please revise your proposal accordingly.

18/10/2025 10:51 pm

Got it, thank you for the suggestion. I will modify these parts.

Haoyu Liu 21/02 3:41 pm

I hope you had a good break. Please schedule a meeting with me to discuss your plan for this semester. I will be available
February 24 (Tuesday) 14:00-16:00
February 25 (Wednesday) 14:00-16:00
February 27 (Friday) 10:30-12:00 14:00-16:00
March 3 (Tuesday) 14:00-16:00
March 4 (Wednesday) 14:00-16:00

21/02 3:42 pm

I am available in February 24 14:00 Tuesday 🙂

Haoyu Liu 21/02 3:46 pm

Time: Feb 24, 2026 02:00 PM Canberra, Melbourne, Sydney

Join from PC, Mac, Linux, iOS or Android:
https://zoom.uts.edu.au/j/84837436888

21/02 3:46 pm

got it

Tuesday, 10 March

10/03 4:32 pm

Due to email file size restrictions, I'm sending you this message via Teams. I wish to present my recent work: I've developed a game demo and recorded a video, which I've included as an attachment. The theme revolves around aerial combat, where players launch missiles to attack enemies ahead and progress through levels. Periodically, players can enter a shop to make purchases. Available items include missile speed and power upgrades, shield items, health regeneration, and increased maximum health. Three enemy types have been designed, with a boss battle occurring every five levels.

Air Combat Game Demo.MP4

Wednesday, 11 March

Haoyu Liu 11/03 2:19 am

Let us schedule a meeting to discuss. I am available most of this week.

11/03 3:26 am

Would Thursday at 3pm be acceptable?

Haoyu Liu 11/03 1:25 pm

Time: Mar 12, 2026 03:00 PM Canberra, Melbourne, Sydney

Join from PC, Mac, Linux, iOS or Android:
https://zoom.uts.edu.au/j/84335971873

11/03 1:43 pm

got it

Monday, 16 March

16/03 9:42 pm

This is my latest updated demo. The main changes include: adding a hard mode, removing the option to purchase health points, adjusting some values, providing detailed descriptions for items, and fixing some bugs discovered during testing.

demo 2.mp4

Wednesday, 18 March

Haoyu Liu 18/03 2:39 am

It looks good!

1. I suggest separating the scoring system from the wealth system; otherwise, there will be a snowball effect, i.e., the higher the score the player gains, the more powerful the player will become.

2. I suggest unifying item prices/costs so that adding attack speed and adding damage are comparable.

3. Please check 5:15 or 5:25 of the demo. Is the lost HP displayed as transparent or as black? I suggest standardizing it.

Thursday, 19 March

19/03 9:44 pm

The scoring system is now in place and can be found in the top-right corner of the interface. The prices for speed and damage items are standardized, starting at 100 yuan and increasing by 25 yuan per level, with each level providing a 10% boost to the effect. I've updated the UI for health loss; it now features a uniform black background.

Friday, 3 April

3/04 4:15 pm

I will set aside some time to complete this report, but I have a few questions before I start:
1. Can I use the current version of the game as-is for this assignment?
2. Is there a specific report template tailored to the content of our game? I checked Canvas, but it doesn't seem to match what I recall being discussed in the meeting. I'd like to ask if there's anything I can use as a reference.
3. Does the final report require uploading the actual code from Coding? I developed the game on the platform, so I might have some difficulty obtaining the code if it's required.

Sunday, 5 April

Haoyu Liu 5/04 9:57 am

1. Yes. The current version of the game is reasonably good.

2. No. Nevertheless, your report should include the essential elements of a typical academic report, such as Introduction and Literature Review sections. The main body of your report may be organized to suit your project, for example:
(1) Describe how you designed and developed the game, presented as the Method section.
(2) Describe how people may play and use the game, presented as the Results section.

3. No.

Research Question Thinking ...
18 KB

Dear Haoyu,

I hope this message finds you well.

Following our previous discussion, I undertook a preliminary analysis of the issues you raised [The distinction between in-game limited-time countdowns and those typically found in e-commerce platforms or restaurants] and consulted several relevant sources. I have compiled these findings into a document, which is attached.

Furthermore, in light of the forthcoming literature review and other assignments, I would appreciate your guidance on how best to approach these tasks and the subsequent schedule.

I extend my sincere gratitude for your continued guidance and support.

Warm regards,
Mingrui Qi
Student ID: 25515192

Hi Mingrui,

I posted very detailed guidelines for the Literature Review on Teams on August 21. Please refer to my post on Teams.

Best,
Haoyu

literature review.docx
23 KB

Dear Haoyu,

I hope this email finds you well.

I'm writing to let you know that I have completed the first draft of my literature review. I have attached the document for your perusal.

Additionally, I have reviewed the assignment instructions regarding the peer review process, but I remain uncertain about the specific timing and operational procedures for the peer review workshop. Would you kindly confirm the exact date, time, and location for the draft exchange and review meeting?

Thank you once again for your guidance and support.

Kind regards,
Mingrui

Hi Mingrui,

The literature review looks good.

The peer review for the literature review is scheduled for the week of 15–19 September. Please check by yourself which specific session you are assigned to.

Best,
Haoyu